FLEX: Full-Body Grasping Without Full-Body Grasps

Purva Tendulkar *Dídac Surís* *Carl Vondrick*

COLUMBIA UNIVERSITY

flex.cs.columbia.edu

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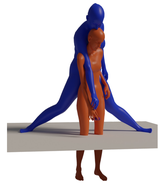
Task and Challenges

Full-body grasping of objects in presence of obstacles.

MoCap Challenges

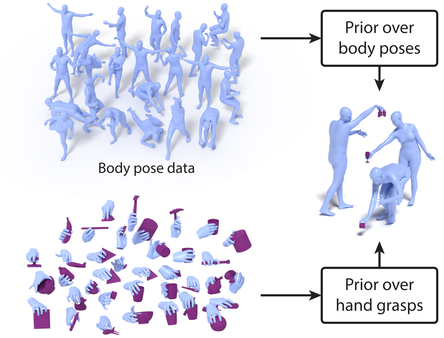


No Generalization

No Obstacle Consideration

Key Idea

- Full-Body Grasp → Full-Body + Hand-Grasp
- No explicit full-body grasps required for training.

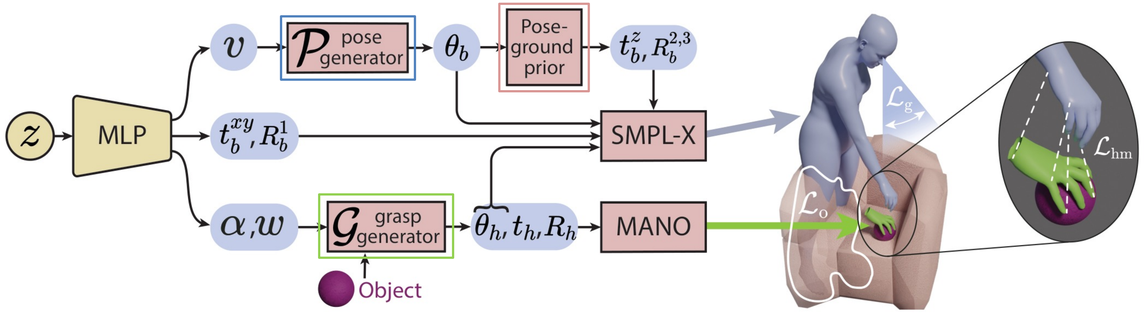


ReplicaGrasp Dataset

Approach

FLEX (**F**ull-Body **L**atent **Ex**ploration) generates a 3D human grasping the desired object, given

1. Pre-trained right-hand grasping model $\mathcal{G}$ that can predict global MANO parameters $\{\boldsymbol{\theta_h}, \boldsymbol{t_h}, \boldsymbol{R_h}\}$
2. Pre-trained full-body pose prior $\mathcal{P}$ that can generate feasible full-body poses $\boldsymbol{\theta_b}$
3. Learnt pose-ground prior which predicts the floor given the human pose $\boldsymbol{\theta_b}$



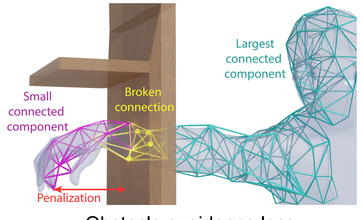
FLEX performs a gradient-based search over hand & body priors to minimize hand-matching and obstacle losses.

Key Insights

- Searching in the latent space of the hand-grasping model $\mathcal{G}$ allows full-body obstacle consideration.

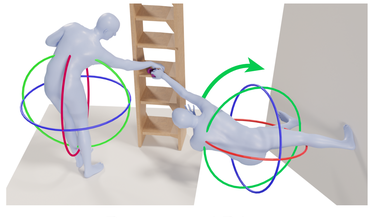
Hand-grasp search

- We penalize *all* vertices in the connected components of the resulting body graph other than the largest one.



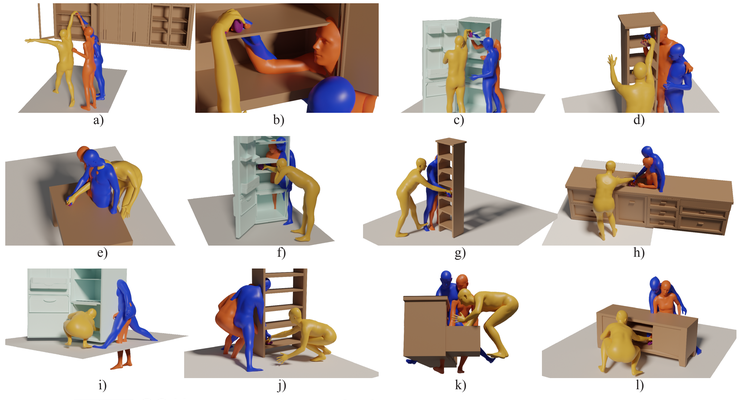
Obstacle avoidance loss

- The ground position can be predicted from the body pose. This removes 2 DoFs from the optimization.



Pose-ground Prior

Comparison Results



Diversity Results

All humans are samples from FLEX